Military College of Signals

National University of Sciences & Technology

Computer & Communication Networks

Submitted to: Lab Engr Alishba Azam

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Lab Report Number	09
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RUBRICS FOR LAB PERFORMANCE

CLO4-PLO5

Name	**Muhammad Sameed khattak**	**Wahaj Shahid goraya**	**Awais Bin Asif**	
R1 Network components				
R2 Demonstration				
R3 Modern tool Usage				

CLO5-PLO9

R1 Contribution/ Group participation				

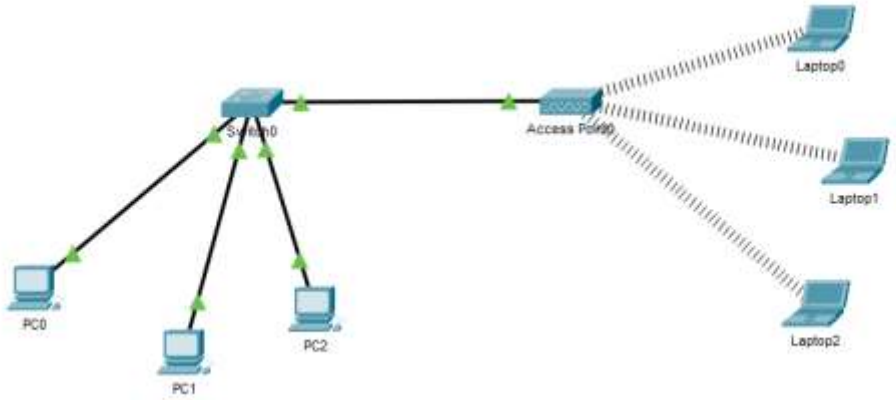
Lab Experiment 09:

LAN using Wireless Access point and Switch and Virtual Local Area Network – Packet Tracer

Objectives:

In this lab we will cover the following

- Creating a point to point network using Ethernet cable
- Creating a local area network using wireless access point
- Creating a local area network using Switch
- Combining both wired and WLAN to have a single network
- Ping the nodes to check the connectivity
- Creating different networks using same switch in Packet Tracer Software.
- Connection of Router and Switch along with implementation of inter VLAN Routing.
- Ping the nodes of different networks to check their connectivity.



In Lab

The network shown in the above diagram has to be established. Its procedure is given below:

1. You are given a TP – Link Access point and you are supposed to connect nodes wirelessly with it to form a W-LAN.
2. First step is to note the IP address, user name and password of access point

which is printed on the back side of AP.

3. By default, IP of AP is 192.168.1.1 and user name, password is admin, admin.
4. Next step is to configure the access point by accessing its graphical user interface.
5. For that, AP is connected directly through a Cross Twisted Pair cable with NIC card of a PC.
6. In order to access the GUI of AP, you are required to establish a network between AP and PC.
7. For that, a static IP must be assign to PC of the same network as that of AP but host ID should be different from that of AP
8. When a network is established between PC and AP, next step is to add the IP of AP in proxy exempted list of PC.

9. This is done using the following method

    - Open internet explorer from PC, then settings then internet options, then connections.
    - From LAN settings, you can see is proxy server is enabled or not. If it is enabled, then go to advanced tab, you will see a block where you can add the addresses for which you don't want to use proxy server. Add IP of AP here.

10. Next step is to open the browser and type the IP of AP, it will ask user name and password to access the GUI.
11. Provide user name and password and GUI of AP will be loaded.
12. AP configuration is given in the reference file which has been provided with this lab manual.
13. Next step is to connect Laptops wirelessly with AP using its SSID and key.
14. Connectivity of systems can be confirmed by pinging the IP's of every connected system provided to it by AP through its DHCP.

15. If ping of all the systems is successful then network has been established successfully.

16. Now connect the provided Switch in order to configure it.

17. Establish a network between Switch and PC as similar to that of AP.

18. Go to browser and type the management IP of Switch (Printed on it), in order to load its GUI. It will ask you a password. Type the default password printed on the Switch, its GUI will be loaded.

19. From here you can see the port status, connection and VLAN status.

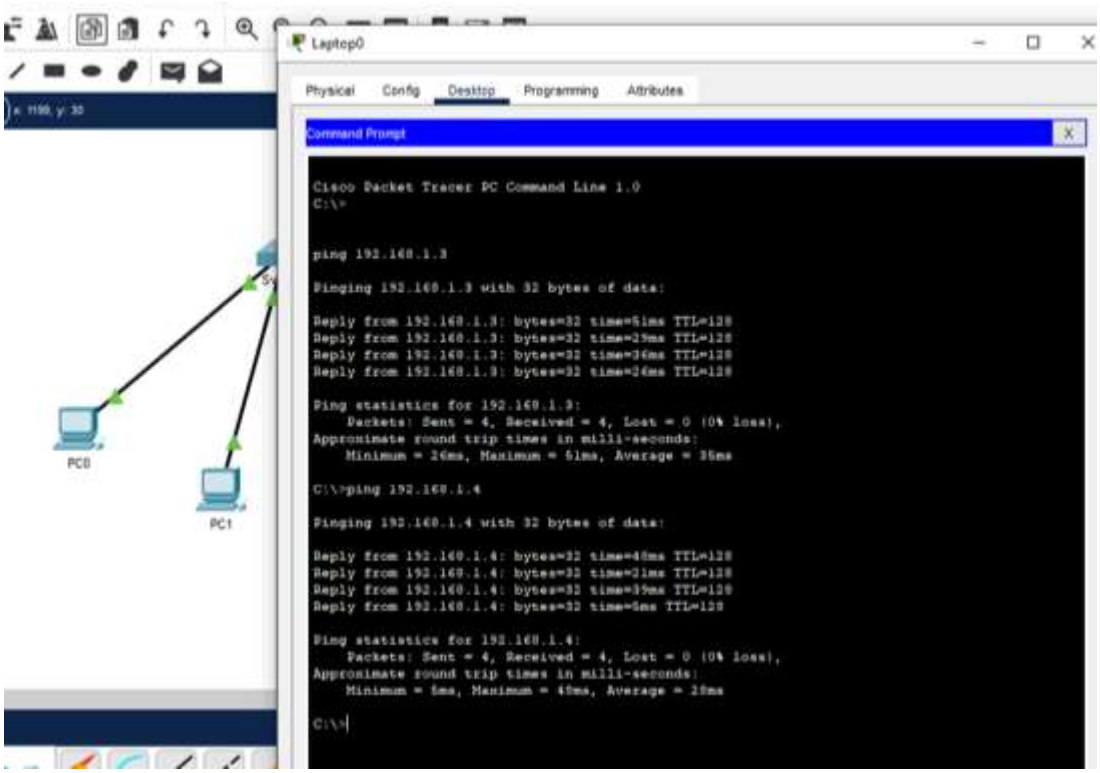
20. Now Connect the AP to one port of switch, remember AP should be on the network of Switch, for this you have to change the IP of AP.

21. Connect PC's with Switch and their IP's should be on same network as that of switch.

22. Wirelessly connect some PC's with AP, and they will be provided IP of same network by DHCP of AP.

23. Final step is to check the connectivity and do sharing of data.

24. Ping all the PC's and if ping is successful, it means that your network is now perfectly fine and data can be shared, any system on network can be accessed.

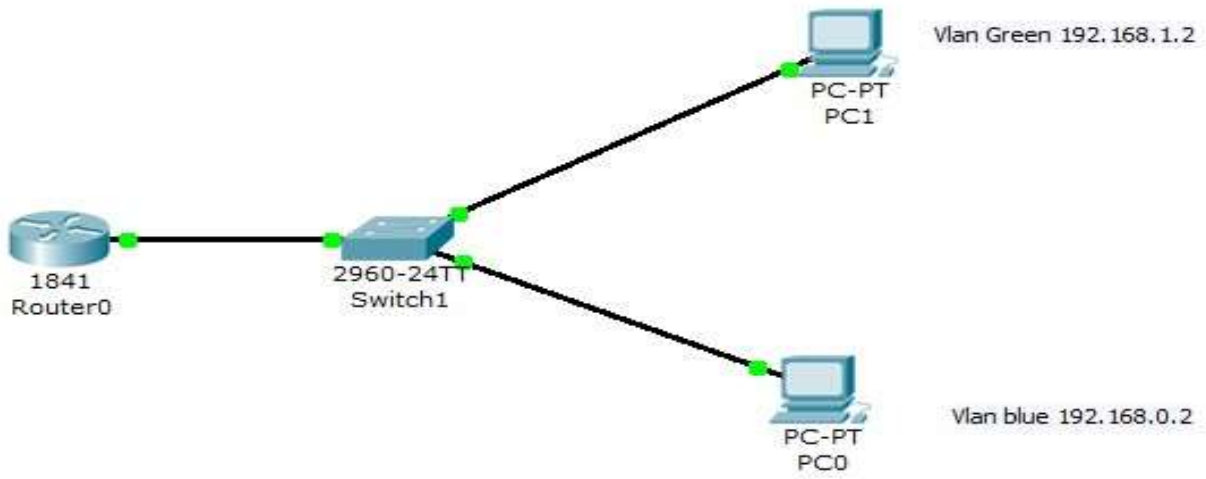


Virtual Local Area Network

Virtual Local Area Network

As shown in fig.2, the above network with 2 VLANs (named Green and blue) is to be established. Its procedure is given below

Pre Lab



Introduction to Virtual LAN

"A Virtual or VLAN is a Logical Grouping of Network Users and Resources connected to administratively defined ports on a switch." To create a computer network, normally all the PCs are connected to a central switch and a star topology is created. The central Switch is a layer 2 device that maintains a Filter/MAC/Forwarding table. In that star topology, Switch receives the frame from a PC and according to its Table, it decides to forward, flood or drop the packet.

However, its functionality is not limited to that only. If we have a Switch and we want to group our PC's connected to it into multiple logical segments, or in other words, **create different networks using same switch**, we create **virtual LANs**. For example, if 6 PCs are connected to a SWITCH, 3 VLANS can be created. Each VLAN will have 2 PCs, thus divided into three logical segments (3 different IP Networks). Physically all the PCs appear to be in a single LOCAL AREA NETWORK but virtually they are 3 in number.

A VLAN allows you to segment a network without needing separate hardware. So, you can have a single physical switch, but multiple different networks connected. Some of the advantages of VLANs are,

- help with network efficiency by reducing extraneous traffic.
- enhance security by creating a virtual boundary around different virtual groups.
- improve bandwidth performance by limiting node-to-node and broadcast traffic.

All the VLANs can be different Networks or different Subnets. So, by having 3 VLANs, there will be 3 broadcast domains (1 for each network) and each VLAN can communicate with nodes of its own group only. Inter VLAN communication is not possible without a router. To have inter VLAN communication, a router is required to be connected with the switch. All the Switch ports that are configured in a VLAN are called **access ports** and others that are connected to routers for carrying the switch data are called **trunk ports**.

In Lab

Procedure:

- Open packet tracer environment and create topology as shown in the figure 1.
- In this scenario, we are creating 2 VLANs, so connect two PC's to a 24 port (2960) switch.
- Set IP addresses PC1: 192.168.1.2
  PC2: 192.168.0.2

  (Note that they are 2 different networks now)

- Next step is to connect a router (1841) to Fast Ethernet port of Switch. Fa0/0 of router is connected to switch.
- Now to configure the switch enter into its CLI mode.
- When its IOS gets loaded, get in the interface mode of the switch by using the following commands.

Switch Configuration:

- **Switch>enable**
- **Switch#configure terminal**
  We are creating 2 VLANs named green and blue, remember to start VLAN creation from VLAN 2 because VLAN 1 is a default VLAN and its name can't be changed.

- **Switch(config)# vlan 2**
- **Switch(config)#name green**
- **Switch(config)#vlan 3**
- **Switch(config)#name blue**
- Now getting to the interface to which PC0 is connected, we have to put Fast Ethernet0/0 port in access mode and in VLAN 2
  - **Switch(config)#int fa0/0**
    (PC1 is connected to interface fa0/0 of switch)

  - **Switch(config-if)#switchport mode access**
  - **Switch(config-if)#switchport access vlan 2**
  - **Switch(config-if)#no shut**
  - **Switch(config-if)#exit**
- Repeat the above procedure by getting to the interface to which PC1 is connected, put Fast Ethernet0/1 port in access mode and in VLAN 3
- Now configure Fa0/3 of Switch in trunk mode which is connected to router.
- **Switch(config)#int fa0/3**

- **Switch(config-if)#Switchport mode trunk**
- Press CTRL^Z to get out of the privilege mode

Router Configuration:

- Now it's time to configure the router so get into CLI mode of router.
- **Router > enable**
- **Router # configure terminal**
- **Router(config) #**
- **Router(config) # int Fa0/0**
- **Router(config-if) # no shut**
- **Router(config-if) # Int f0/0.1**
- **Router(config-subif) # encapsulation dot1q 2**
- **Router(config-subif) # Ip address 192.168.1.1 255.255.255.0**

  (I have attached this sub interface in VLAN 2 and now I am assigning it an IP (192.168.1.1) so this will act as gateway for PC1)

- **Router(config-if)#int f0/0.2**
- **Router(config-subif) # encapsulation dot1q 3**

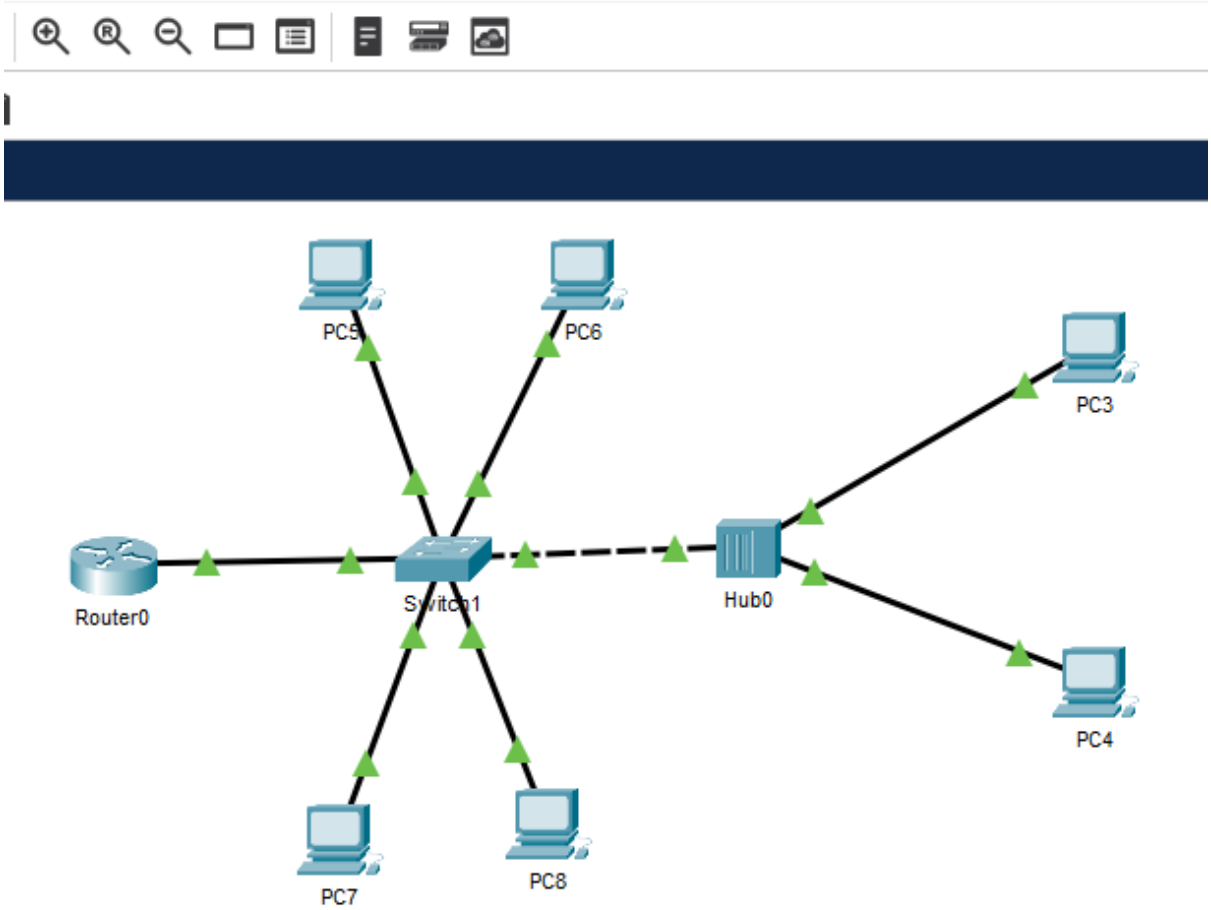
  (I have attached this sub interface in VLAN 3 and now I am assigning it an IP (192.168.0.1) so this will act as gateway for PC0)

- **Router(config-subif) # ip address 192.168.0.1 255.255.255.0**
- **Router(config-subif) # exit**
- Press CTRL^Z to get out of privileged mode.
- Now you can see a green light on router interface which means it has been configured.
- Again, select CLI of switch and in privileged mode (mode activated by enable), type

  Show vlan And **Show int trunk**

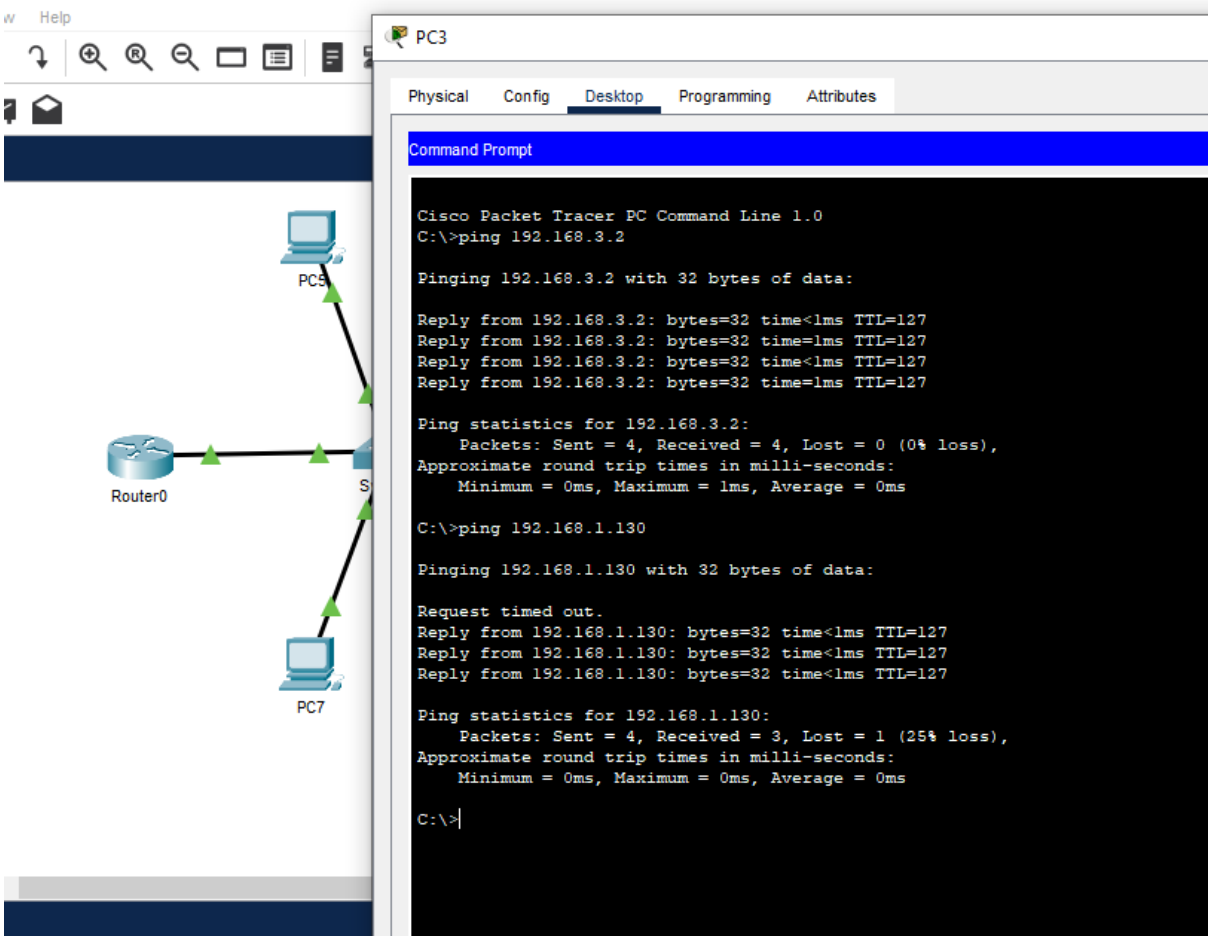
  From this you can verify VLAN and TRUNK ports.

- Now ping the gateway and both PCs, if successful then you have successfully created 2 VLANs.



Post Lab

2. Your Task is to create the above topology of VLANs and create 3 VLANs, first by 3 different networks and then by 3 subnets of same network. Configure Inter VLAN Routing.


Help
PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.2
Pinging 192.168.3.2 with 32 bytes of data:
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.3.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.130
Pinging 192.168.1.130 with 32 bytes of data:
Request timed out.
Reply from 192.168.1.130: bytes=32 time<1ms TTL=127
Reply from 192.168.1.130: bytes=32 time<1ms TTL=127
Reply from 192.168.1.130: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.1.130:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
PC5
Router0
PC7

PC5

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Request timed out.
Reply from 192.168.1.2: bytes=32 time=1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=10ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=21ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 21ms, Average = 10ms

C:\>
```

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#interface fa0/0.10
Router(config-subif)#ip address 192.168.1.1 255.255.255.192
Router(config-subif)#exit
Router(config)#
Router(config)#interface fa0/0.20
Router(config-subif)#ip address 192.168.1.65 255.255.255.192
Router(config-subif)#exit
Router(config)#
Router(config)#interface fa0/0.30
Router(config-subif)#ip address 192.168.1.129 255.255.255.192
Router(config-subif)#exit
Router(config)#
Router(config)#
```

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
10   VLAN1                            active    Fa0/2
20   VLAN2                            active    Fa0/3, Fa0/4
30   VLAN3                            active    Fa0/5, Fa0/6
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```